Sampling Method: Accept-Reject Sampling

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Main Problems with Direct Sampling

- Standard sampling methods may not work well in certain cases. For example, the Inverse Transform Sampling method is ineffective when the distribution lacks an easy-to-calculate inverse cumulative distribution function (CDF), such as in the case of the Gamma distribution.
- In some situations, the numerator of a probability density is known, but the denominator is unknown. For example, in Bayesian inference:

$$P(\theta \mid \mathbf{x}) \propto P(\mathbf{x} \mid \theta)P(\theta),$$

where the posterior $P(\theta \mid \mathbf{x})$ is proportional to the product of the likelihood $P(\mathbf{x} \mid \theta)$ and the prior $P(\theta)$.
Calculating the exact value of the following integral, which serves as a normalizing constant, can be very challenging:

$$P(\mathbf{x}) = \int P(\mathbf{x} \mid \theta)P(\theta)\, d\theta.$$

Main Problems with Direct Sampling

A specific example

Suppose we have a probability density function (PDF) $p(s)$, the numerator of $p(s)$ is known as $f(s)$ and the denominator is unknown which is NC

$$f(s) = \begin{cases} e^{-\frac{s}{2}} + e^{-s^2} & \text{if } s \geq 0, \\ e^{\frac{s}{3}} + e^{s^3} & \text{if } s < 0. \end{cases} \tag{1}$$

$$p(s) = \frac{f(s)}{NC} \tag{2}$$

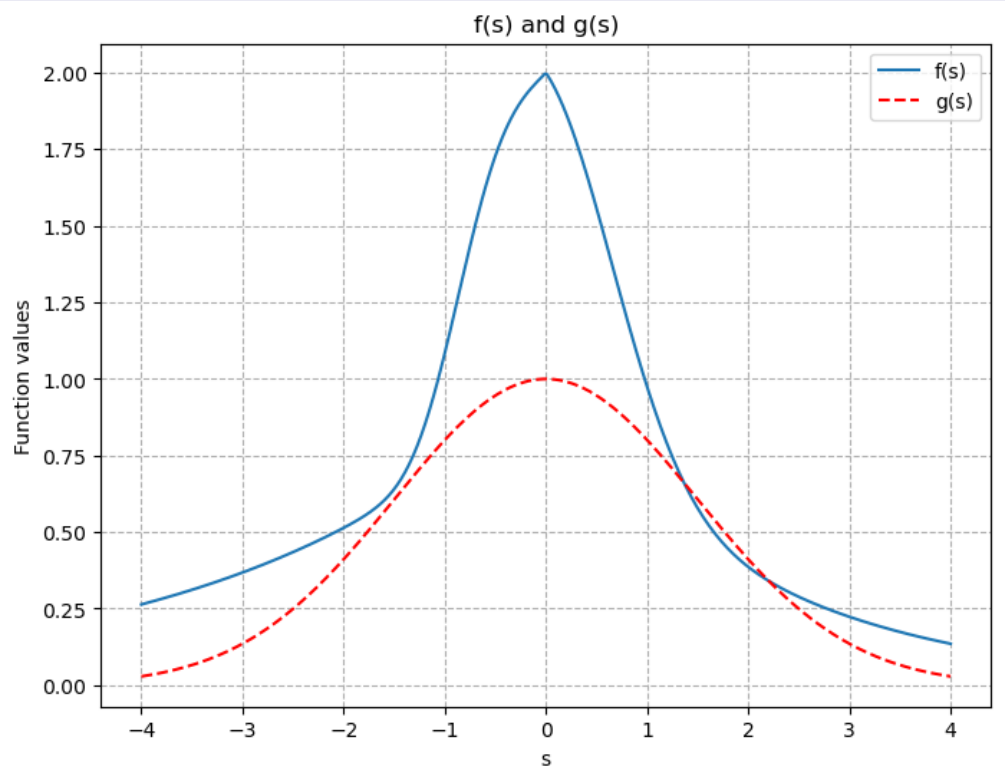
It is difficult to calculate the denominator.

$$NC = \int_{-\infty}^{\infty} f(s)\, ds$$

How do we sample from $p(s)$?

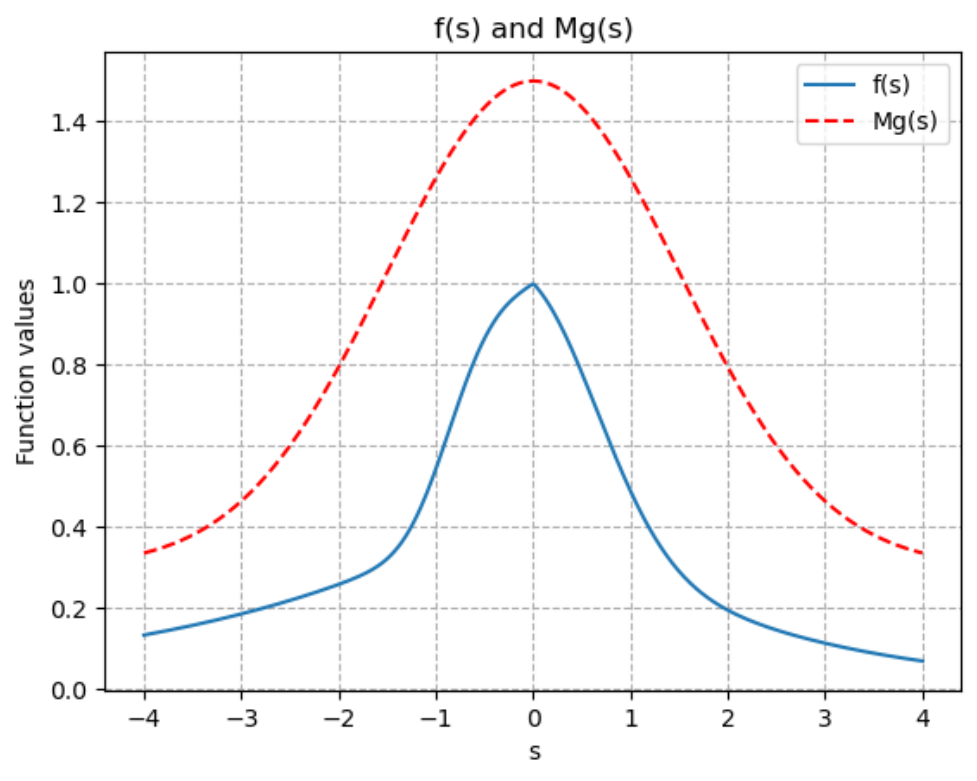
How Accept-Reject Sampling Works

Step 1: Choose $g(s)$ which is "Close" to $p(s)$ and Easy to Sample.



How Accept-Reject Sampling Works

Step 2: Make Sure the line of g(s) above the line of f(s)
– Scale $g(s)$ by M.



How Accept-Reject Sampling Works

Step 3: Accept or Reject a Sample with Probability $\frac{f(s)}{M \cdot g(s)}$.

Keep drawing samples from $g(s)$, and for each sample, decide to accept or reject it based on the acceptance probability $\frac{f(s)}{M \cdot g(s)}$. Specifically:

- Generate a random value u from the Uniform(0,1).
- If $u \leq \frac{f(s)}{M \cdot g(s)}$, accept the sample s; otherwise, reject it.

How Accept-Reject Sampling Works

Algorithm

1. Generate s from $g(s)$.
2. Generate u from Uniform(0,1), $u \sim \mathcal{U}(0,1)$.
3. If $u \leq \frac{f(s)}{M \cdot g(s)}$, keep s, else go back to 1.

Why Accept-Reject Sampling Works

Why does following this algorithm yield samples from the target distribution $p(s)$? By Bayes Theorem, the pdf for accepted samples can be checked as follows:

$$p\left(s \mid u \leq \frac{f(s)}{M \cdot g(s)}\right) = \frac{p\left(u \leq \frac{f(s)}{M \cdot g(s)} \mid s\right) g(s)}{\int p\left(u \leq \frac{f(s)}{M \cdot g(s)} \mid s\right) g(s)\, ds}$$

$$= \frac{\frac{f(s)}{M \cdot g(s)} \cdot g(s)}{\int \frac{f(s)}{M}\, ds}$$

$$= \frac{NC \cdot p(s)}{NC \cdot \int p(s)\, ds} = p(s)$$

Why Accept-Reject Sampling Works

A further question may arise as to why we need to sample u from a uniform distribution. This requirement is due to the Fundamental Theorem of Simulation. More details are available in Robert and Casella (2004), Page 47.

Theorem 2.15: Fundamental Theorem of Simulation

Simulating $X \sim f(x)$ is equivalent to simulating

$$(X, U) \sim \mathcal{U}\{(x, u) : 0 < u < f(x)\}.$$

Corollary 2.17:

Let $X \sim f(x)$ and let $g(x)$ be a density function that satisfies $f(x) \leq Mg(x)$ for some constant $M \geq 1$. Then, to simulate $X \sim f$, it is sufficient to generate

$$Y \sim g \quad \text{and} \quad U \mid Y = y \sim \mathcal{U}(0, Mg(y)),$$

until $0 < u < f(y)$.

Key Considerations

- $g(s)$ should 'look like' $f(s)$ to avoid wasting simulations.
- While it is not necessary, it is common practice to set $u \sim \mathcal{U}(0,1)$, so that we can interpret $\frac{f(s)}{M \cdot g(s)}$ as the acceptance "probability".

Potential Challenges

- What happens if there is a spike in $f(s)$?

  We would need to use an extremely large M to scale $g(s)$ up enough to exceed this spike. As a result, the acceptance probability $\frac{f(s)}{M \cdot g(s)}$ approaches zero, making it take a very long time to obtain the desired number of samples.
- Is it always possible to scale $g(s)$ so that it exceeds $f(s)$? The answer can be explored through the following code.

Code and Bibliography

Code

You can find an example code for the Accept-Reject Sampling method in the file MCMC Experiments.ipynb at the following link:
https://github.com/ritvikmath/YouTubeVideoCode

Bibliography

Robert, C. P. and Casella, G. (2004). *Monte Carlo statistical methods*. Springer texts in statistics. Springer, New York ;, second edition. edition.